



Research paper

Antimicrobial resistance profiling of bacteria isolated from wastewater and samples of pharmaceutical industries in South India

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ABSTRACT

The study was aimed to determine the phenotypic and genotypic antimicrobial resistance in the isolated bacteria from the influent (25), effluent (15), surface and ground water samples (15) surrounding the pharmaceutical industries located in south India. From 55 samples, 48 isolates of 10 different bacteria were obtained. The identified bacterial isolates were viz. *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Enterobacter aerogenes*, *Corynebacterium* sp., *Acinetobacter* sp., *Aeromonas punctata*, *Ralstonia picketti*, *Staphylococcus aureus*, *Stenotrophomonas maltophilia*, and *Citrobacter freundii*. The phenotypic profile of resistance through antibiotic susceptibility test was carried out against sixteen different antibiotics. Standard PCR technique was used for the detection of 12 resistance genes encoding carbapenems, quinolone, aminoglycoside, β -lactam belonging *blaOXA-58*, *blaOXA-22*, *qnrA*, *qnrB*, *aac(6)-Ib-cr*, *aac(3)-XI*, *mecA*, *qepA*, *aadB*, *blaVIM*, *blaOXA-48* and *blaNDM*. *Pseudomonas aeruginosa* (1: TN/1/2020) showed presence of 3 resistance genes. *qnrB* (489 bp) gene was present in maximum of 7 isolates while *blaVIM* (196 bp) gene was present in 6 isolates. The resistance genes *blaNDM* (621 bp) was present in three different isolates; *aac(X):6-Ib-cr* (482 bp), *qepA* (495 bp), *aadB* (500 bp), *blaOXA-58* (843 bp) resistant genes were present in two different isolates each among the bacterial isolates obtained in this study. In phenotypic resistance profiling by AST method, out of 16 antibiotics tested, 14 showed resistance. Similarly, in genotypic resistance profiling, among 12 resistance genes tested, a maximum of three resistance genes were noticed in *Pseudomonas aeruginosa*. There were positive and negative correlations observed between phenotypic and genotypic resistance among different antibiotics and their resistance genes indicating the variations in the resistance gene expression.

1. Introduction

Antimicrobial resistance has a significant effect on public health, resulting in increasing health care expenses and mortality rates. According to the literature, around 4.5 million people die each year as a consequence of antibiotic resistant pathogen infections, with up to 10 million fatalities estimated yearly, with 2 million expected to occur in India, which is expected to be the leading cause of death globally by 2050 (Mohsen Naghavi, 2022). Although an increasingly serious issue that is emerging concerns antibiotic resistance. The majority of drugs used to treat chronic infections in both humans and animals are often antibiotics, in addition to being a component of animal feed and

fisheries production that promotes growth (Rueanghiran et al., 2022).

It has been noticed that antibiotic resistance is detected not only in pathogenic bacteria but also in environmental living creatures in terrestrial and aquatic environments. Polluted environments exhibit higher concentrations of resistant bacteria than unpolluted ecosystems are, demonstrating that human activity has had a major role in the increase of resistant bacteria in the environment (Obayiuwana et al., 2018; Mahmood et al., 2019).

Antibiotics cause resistant bacteria to adapt to their environments and act as vectors for the spread of antibiotic resistance due to their ability to kill or inhibit the growth of susceptible bacteria. The major threat to public health is the transmission of resistance genes from

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environmental bacteria to human pathogens (Bielen et al., 2017; Ngidi et al., 2020).

Drug resistance in bacteria has recently been found to have the potential to promote their development and dissemination. It not only makes things more complicated, but it also contributes to issues with antibiotic resistance spreading across the ecosystem and multidrug resistance (MDR). The existence of MDR strains in water sources throughout the animal production process has been confirmed in multiple surveys, which results in the continuous spread of resistance genes across bacterial populations. Because of Antibiotic Resistance Genes (ARGs) genetic elements as the existence of components that allow for the absorption of design features into the environment via the discharge of organic wastes, concerns about public health emerge because it may pose risks to both human health and the environment (Xu et al., 2020; Reinthaler et al., 2010; Dawangpa et al., 2021; Tong et al., 2021; Posebón et al., 2021).

Several antibiotic resistance genes found in chromosomes or plasmids lead to antimicrobial resistant bacteria, which can transfer their resistance traits to non-resistant bacteria. On the other hand, horizontal gene transfer allows resistance genes to spread among resistant bacteria, creating multidrug-resistant bacteria. One important process that contributes to the quick dissemination of antibiotic resistance genes in environmental bacteria is horizontal gene transfer. Resistance bacteria are of pathogenic and non-pathogenic nature and can affect people and animals by exposure to contaminated water or environment. When an infection occurs with a resistant bacterium in human or animal, treatment efficiency reduces which may lead to death (Da Costa et al., 2013; Atlaw et al., 2021).

Pharmaceutical influent refers to the incoming wastewater that flows into a treatment facility. This water includes waste from hospitals, pharmaceutical manufacturing companies, and households where medications are used. It often contains drug residues, including antibiotics, as well as other chemicals and biological materials.

Pharmaceutical effluent is the treated wastewater that leaves a treatment facility. Although the treatment process is designed to remove contaminants, some pharmaceutical compounds and antibiotic resistance genes (ARGs) may not be completely broken down. Consequently, traces of drug residues and ARGs can remain in the effluent, which is often released into surface water bodies such as rivers and lakes, or directly into groundwater.

The study was planned to investigate pharmaceutical industries involved in antibiotic production. It focuses on whether antibiotic residues from these industries enter the environment through influent and effluent, potentially contaminating environmental sources such as bore wells, rivers, and groundwater. This research analyzes and examines the extent of such contamination.

This study was carried out to determine the phenotypic and genotypic pattern of antimicrobial resistance in the isolated bacteria from pharmaceutical influents, effluents, nearby surface and ground water in south India.

2. Materials and methods

2.1. Source materials

A total of 55 water samples were collected, among which 25 samples were from influent, 15 samples were from effluent and 15 samples were from surface, ground water samples were collected from several pharmaceutical companies that manufacture antibiotics in south Indian states like Tamil Nadu (TN), Andhra Pradesh (AP), Telangana (TS), Kerala (KL), and Karnataka (KA). Bacterial isolates were identified using VITEK 2 Compact and VITEK MS (BioMérieux, India (P) Ltd., New Delhi) for further confirmation. Forty-eight isolates were obtained viz. *Klebsiella pneumoniae* (7), *Pseudomonas aeruginosa* (9), *Enterobacter aerogenes* (6), *Corynebacterium sp* (1), *Acinetobacter sp.* (2), *Aeromonas punctata* (2), *Ralstonia picketti* (1), *Staphylococcus aureus* (16), *Stenotrophomonas*

maltophilia (2), *Citrobacter freundii* (2).

2.2. Phenotypic assessment of antibiotic resistance

The phenotypic pattern of resistant and susceptibility antibiotic against *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Enterobacter aerogenes*, *Corynebacterium sp.*, *Acinetobacter sp.*, *Aeromonas punctata*, *Ralstonia picketti*, *Staphylococcus aureus*, *Stenotrophomonas maltophilia* and *Citrobacter freundii* bacterial isolates obtained were investigated using the disk diffusion method on Muller-Hinton agar (Merck, Germany). Clinical Laboratory Standard Institute (CLSI) guidelines were followed in this study (Baseri et al., 2021). Diverse kinds of antibiotic agents including amino glycosides (Gentamicin and amikacin), cyclic polypeptide (bacitracin), macrolides (erythromycin), fluoroquinolones (ciprofloxacin), lincosamide (clindamycin), β -Lactam derivatives (cephalosporin - ceftazidime, cefoxitin, cefotaxime, Ceftazepzone / Sulbactam, Piperacillin / Tazobactam, Ampicillin, Penicillin), glycopeptides (Vancomycin), carbapenems (Imipenem) and oxazolidinone (Linezolid) were used for the test (Oxoid, UK).

2.3. Genotypic assessment of antibiotic resistance

Standard PCR technique was employed for detection of genes encoding resistance to carbapenems, quinolone, aminoglycoside, β -lactam belonging *bla_{OXA-58}*, *bla_{OXA-22}*, *qnrA*, *qnrB*, *aac(6)-Ib-cr*, *aac(3)-XI*, *mecA*, *qepA*, *aadB*, *bla_{VIM}*, *bla_{OXA-48}* and *bla_{NDM}*. PCR was optimized for the detection of all these 12 antibiotic resistance genes among Gram positive and Gram negative bacterial isolates because of the possibility of transfer of resistance genes across the group and were amplified by single and/or multiplex polymerase chain reaction (mPCR) (Poirel et al., 2005; Hosu et al., 2021; Girlich et al., 2004; Plante et al., 2003; Nowrotek et al., 2021; Robicsek et al., 2006; Park et al., 2006; Galimand et al., 2015; Abdolmaleki et al., 2019; Poirel et al., 2011) following the protocol shown in Table 1 and Fig. S1, S2, S3, S4, S5, S6, S7, S8, S9, S10, S11, S12, S13, S14, S15, S16.

2.4. Preparation of template DNA

A single colony of each organism from the respective agar plate was inoculated into 1.5 ml of Luria-Bertani broth (HiMedia laboratories, Mumbai.), and incubated overnight at 37 °C. Cells from the overnight culture were centrifuged at 12000 rpm for 10 min. Supernatant was decanted and the pellet was re-suspended in 200 μ l of sterile distilled water. The cells were lysed by heating at 100 °C for 10 min. 2 μ l of supernatant of the total sample, was used as the source of the template for amplification. The DNA was stored at -20 °C for PCR analysis. The purity (A_{260}/A_{280}) and concentration of extracted DNA were then checked (NanoDrop, Thermo Scientific, USA). The primers used were synthesized (Eurofins genomics Pvt. Ltd., Bangalore, India) and the PCR master mix (Red dye master mix, Amplicon, Denmark) was used to set the PCR reaction. For optimization of PCR, previously well characterized strains that had been recovered from different parts of the world were used as positive control and negative control (double sterilized dist. water). A band with the expected size in comparison with the positive control was considered positive. The PCR amplicons were assessed using 1 % agarose gel electrophoresis stained with ethidium bromide (0.5 μ g/ml) (Thermo Fisher Scientific, Germany) (Shivakumaraswamy et al., 2019; Akinbami et al., 2018; Mukherjee et al., 2021).

2.5. Bio-informatics analysis

Antimicrobial resistance profile (phenotypic and genotypic) was converted into binary coding (positive/resistant = 1, negative/ susceptible = 0). Hierarchical clustering was designed and analysed using Displayr software (<https://app.displayr.com/>). The correlation matrix between phenotypic and genotypic antimicrobial resistance was created

Table 1

Table showing details on the list of antibiotic resistant genes, primer sequences, PCR conditions and amplicon.

Target gene	Primer Sequence (5' - 3')	Conditions						Amplicon size (bp)
		Initial denaturation	Denaturation	Annealing	Extension	Final extension	Cycles	
blaOXA-58 (F)	AAG TAT TGG GGC TTG TGC TG	5 min at 94°C	30 s at 94 °C	55.8°C at 30s	72°C for 45 s	5 min at 72C	25	843
blaOXA-58 (R)	CCC CTC TGC GCT CAT CAT AC							
blaVIM (F)	TGCGCTTCGGTCCAGTAGA	5 min at 94°C	45 s at 94°C	60°C at 45 s	72°C for 45 s	7 min at 72C	45	196
blaVIM (R)	TGACGGGACGTATACAACCAGAT							
bla oxa-22 (F)	GATCGATGATGAAACGCCGC	5 min at 94°C	30 s at 94°C	54.4°C at 60s	72°C for 60s	1 min at 72C	40	769
bla oxa-22 (R)	TTCCGACAGGCGAGCAAACAG							
aadB (F)	CCGCAGCTAGAAATTTTG	5 min at 94°C	45 s at 94°C	58.5°C at 45 s	72°C for 50s	5 min at 72C	30	500
aadB (R)	AGGTTGAGGTCTTGCGT							
qnrA (F)	ATTTCTCACGCCAGGATTTG-3'	7 min at 94°C	45 s at 94°C	53.4°C at 40s	72°C for 60s	5 min at 72C	35	516
qnrA (R)	GATCGGCAAAGGTTAGGTCA							
qnrB (F)	GATCGTGAAAGCCAGAAAGG	7 min at 94°C	45 s at 94°C	52.4°C at 40s	72°C for 60s	5 min at 72C	35	489
qnrB (R)	ACGATGCCTGGTAGTTGTCC							
aac(6)-Ib-cr (F)	TTGCGATGCTCTATGAGTGGCTA	7 min at 94°C	45 s at 94°C	56.7°C at 40s	72°C for 60s	5 min at 72C	35	482
aac(6)-Ib-cr (R)	CTCGAATGCCTGGCGTGT							
aac(3)-XI (F)	GCGGCTACGTAAAGTCC	5 min at 94°C	60 s at 95°C	56.4°C at 60s	72°C for 60s	7 min at 72C	30	400
aac(3)-XI (R)	CCGCCGAGCCGTTTCAT							
mec (F)	ACGAGTAGATGCTCAATATAA	3 min at 94°C	30 s at 94°C	59.4°C at 30s	72°C for 45 s	7 min at 72C	30	310
mec (R)	CTTAGTTCTTTAGCGATTGC							
qepA (F)	CCAGCTCGGCACTTGATAC	3 min at 94°C	45 s at 94°C	56.4°C at 45 s	72°C for 45 s	5 min at 72C	34	570
qepA (R)	ATGCTCGCCTTCCAGAAA							
bla _{OXA-48} (F)	GCGTGGTTAAGGATGAACAC	5 min at 94°C	30 s at 94°C	57°C at 60s	72°C for 50s	5 min at 72C	36	438
bla _{OXA-48} (R)	CATCAAGTTCAACCCAACCG							
bla _{NDM} (F)	GGTTTGGCGATCTGGTTTC	5 min at 94°C	30 s at 94°C	57°C at 60s	72°C for 50s	5 min at 72C	36	621
bla _{NDM} (R)	CGGAATGCGCAGCACCAG							

and analysed using statistical tools for high throughput data analysis online software (www.sthda.com).

3. Results

Antibiotics like Imipenem (IPM); Ceftazidime (CAZ); Ceftaperazone/Sulbactam (CFS); Piperacillin/Tazobactam (PTZ); Amikacin (AK); Ciprofloxacin (CIP); Cefotaxime (CTX) are applicable for *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Enterobacter aerogenes*, *Acinetobacter* sp., *Aeromonas punctata*, *Ralstonia picketti*, *Stenotrophomonas maltophilia*, *Citrobacter freundii* and CIP, Erythromycin (E); Ampicillin (AMP); Gentamicin (GEN); Bacitracin (B); Vancomycin (VA); Clindamycin (CD); Linezolid (LZ); Penicillin (P); Cefoxitin (CX) applicable for *Staphylococcus aureus*, *Corynebacterium* sp. These were used in the phenotypic testing of resistance and susceptibility. In this study, 12 prevalent antibiotic resistance genes encoding for carbapenems, quinoline, aminoglycoside, β-lactam groups viz. bla_{OXA-58}, bla_{OXA-22}, qnrA, qnrB, aac(6)-Ib-cr, aac (3)-XI, mec A, qepA, aadB, bla_{VIM}, bla_{OXA-48} and bla_{NDM} were screened among Gram positive and Gram negative bacterial isolates because of the possibility of transfer of resistance genes across the groups. For bio-informatic analysis, resistance is denoted as 1 in the place of R and susceptibility denoted as 0 in the place of S (Source: Standards for antibiotic disk susceptibility test. Clinical and Laboratory Standard Institutes (CLSI)).

3.1. Phenotypic resistance of antimicrobials in Gram -ve isolates

Almost 24 Gram negative isolates had resistance to all the 6 antibiotics screened in the study viz. PTZ, CFS, CAZ, CIP, IPM and AK. Eight isolates viz. *Pseudomonas aeruginosa* (3 Nos.), *Klebsiella pneumoniae* (1 No.), *Enterobacter aerogenes* (2 Nos.) were susceptible to all the 6 antibiotics tested viz. PTZ, CFS, CAZ, CIP, IPM and AK except AK for the isolate *P. aeruginosa* (1 No.). Isolate *Klebsiella pneumoniae* (1 No.) was susceptible to antibiotics IPM and AK. Isolate *P. aeruginosa* (1 No.) was susceptible to CIP and AK. Another 5 isolates viz. *Acinetobacter* sp. (2 Nos.), *Klebsiella pneumoniae* (2 Nos.) and *P. aeruginosa* (1 No.) were susceptible to Amikacin. *Aeromonas Punctate* (2 Nos.), *Citrobacter freundii* (2 Nos.); *Stenotrophomonas maltophilia* (1 No.), *Ralstonia picketti*

(1 No.) were resistance to all the 6 antibiotics tested (Table 2 and Fig. 1).

3.2. Phenotypic resistance of antimicrobials in Gram + ve isolates

All the isolates were resistant to 8 (CTX, CX, CD, B, GEN, P, AMP, E) out of 10 antibiotics tested in this study. Linezolid and Vancomycin were the only two antibiotics to which all the 17 isolates of bacteria were susceptible. *S.aureus* were susceptible to erythromycin (1 No.), *Corynebacterium* sp. (1 No.). *Corynebacterium* sp. was susceptible to 50 % of the antibiotics tested viz. GEN, P, AMP, E, LZ, VA (Table 3 and Fig. 1).

3.3. Genotypic resistance of antibiotic resistant genes (ARGs) in Gram negative bacteria

Thirty-one Gram negative isolates were also screened for all the 12 ARGs in this study. Six of *Pseudomonas aeruginosa* isolates had bla_{VIM} ARG. qnrA ARG was identified in 2 *Pseudomonas aeruginosa*, 2 *Acinetobacter* sp., *Klebsiella pneumoniae* (1 No.) whereas qnrB ARG was present in 4 isolates of *Enterobacter aerogenes*. Two of *Pseudomonas aeruginosa* isolates, one *Klebsiella pneumoniae* had bla_{NDM} ARG. aac(6)-Ib-cr ARG was present in two of *Aeromonas punctate* isolates. Another ARG – qepA was present in two *Citrobacter freundii* isolates.

Similarly, two *Stenotrophomonas maltophilia* isolates (44: KA/1/2022, 43: KA/GW/2022) had aadB ARG. ARGs bla_{OXA-58}, bla_{OXA-22} were present in *Enterobacter aerogenes* (18: KA/E/2021) and *Ralstonia picketti* (26: AP/GW/2021) respectively. The remaining ARG aac (3)-XI were not present in any of the 31 Gram negative isolates (Table 4 and Fig. 2).

3.4. Genotypic resistance of antibiotic resistant genes (ARGs) in Gram positive bacteria

In this study, 12 antibiotic resistance genes encoding for carbapenems, quinoline, aminoglycoside, β-lactam resistance viz. bla_{OXA-58}, bla_{OXA-22}, qnrA, qnrB, aac(6)-Ib-cr, aac (3)-XI, mec A, qepA, aadB, bla_{VIM}, bla_{OXA-48} and bla_{NDM} were screened. Only three isolates had ARGs viz. *Corynebacterium* sp. (1 No.) with aac (3)-XI ARG, *Staphylococcus aureus* (1 No.) with mecA and *Staphylococcus aureus* (1 No.) with qnrB (Table 5 and Fig. 2).

Table 2

Phenotypic resistant/susceptible pattern of various isolates obtained from the Waste water, Surface water and Ground water samples.

Organism/ Sample ID/ Year	Gram staining	Antibiotics (Resistant/ Sensitive)					
		IPM	CAZ	CFS	PTZ	AK	CIP
<i>Pseudomonas aeruginosa</i> (1:TN/1/2020)	G -ve	S	S	S	S	S	S
<i>Klebsiella pneumoniae</i> (2: AP/GW/2020)	G -ve	S	S	S	S	S	S
<i>Enterobacter aerogenes</i> (2: AP/GW/2020)	G -ve	S	S	S	S	S	S
<i>Klebsiella pneumoniae</i> (3: TS/GW/2020)	G -ve	R	R	R	R	S	R
<i>Enterobacter aerogenes</i> (3: TS/GW/2020)	G -ve	S	S	S	S	S	S
<i>Pseudomonas aeruginosa</i> (7:TS/1/2020)	G -ve	S	S	S	S	S	S
<i>Acinetobacter</i> (8:KA/1/2020)	G -ve	R	R	R	R	S	R
<i>Pseudomonas aeruginosa</i> (9:TS/GW/2020)	G -ve	S	S	S	S	S	S
<i>Acinetobacter</i> (10:TS/1/2020)	G -ve	R	R	R	R	S	R
<i>Pseudomonas aeruginosa</i> (11:TS/1/2020)	G -ve	S	S	S	S	R	S
<i>Pseudomonas aeruginosa</i> (12:TS/1/2021)	G -ve	R	R	R	R	S	S
<i>Enterobacter aerogenes</i> (15:KA/1/2021)	G -ve	R	R	R	R	R	R
<i>Aeromonas Punctate</i> (15: KA/1/2021)	G -ve	R	R	R	R	R	R
<i>Klebsiella pneumoniae</i> &(16:KA/1/2021)	G -ve	R	R	R	R	R	R
<i>Aeromonas Punctate</i> (16: KA/1/2021)	G -ve	R	R	R	R	R	R
<i>Klebsiella pneumoniae</i> (17: KA/GW/2021)	G -ve	R	R	R	R	R	R
<i>Enterobacter aerogenes</i> (18:KA/E/2021)	G -ve	R	R	R	R	R	R
<i>Ralstonia Picketti</i> (26:AP/ GW/2021)	G -ve	R	R	R	R	R	R
<i>Pseudomonas aeruginosa</i> (34:KL/GW/2021)	G -ve	R	R	R	R	R	R
<i>Pseudomonas aeruginosa</i> (35:TS/GW/2021)	G -ve	R	R	R	R	R	R
<i>Enterobacter aerogenes</i> (36:TS/1/2021)	G -ve	R	R	R	R	R	R
<i>Citrobacter freundii</i> (37: TS/1/2021)	G -ve	R	R	R	R	R	R
<i>Citrobacter freundii</i> (38: KA/1/2021)	G -ve	R	R	R	R	R	R
<i>Enterobacter aerogenes</i> (38:KA/1/2021)	G -ve	R	R	R	R	R	R
<i>Stenotrophomonas maltophilii</i> (43:KA/ GW/2022)	G -ve	R	R	R	R	R	R
<i>Stenotrophomonas maltophilii</i> (44:KA/1/2022)	G -ve	R	R	R	R	R	R
<i>Pseudomonas aeruginosa</i> (44:KA/1/2022)	G -ve	R	R	R	R	R	R
<i>Klebsiella pneumoniae</i> (45: KA/1/2022)	G -ve	R	R	R	R	R	R
<i>Klebsiella pneumoniae</i> (51: AP/1/2022)	G -ve	R	R	R	R	S	R
<i>Klebsiella pneumoniae</i> (52: AP/1/2022)	G -ve	S	R	R	R	S	R
<i>Pseudomonas aeruginosa</i> (53:TN/GW/2022)	G -ve	R	R	R	R	S	R

The names of the isolate, for e.g., 37: TS/1/2021 represents Sample number (37): State Name (TS)/Source of Water (I)/Year of collection (2021). The full-form of the abbreviations used are as follows, States from where samples were collected (TN – Tamil Nadu, TS – Telangana State, KL – Kerala, KA – Karnataka, AP – Andhra Pradesh), Source of Water (I – Influent, E – Effluent, GW – Ground Water).

3.5. Correlation matrix analysis of Gram + ve isolates

Among the Gram +ve isolate's phenotypic and genotypic resistance comparison, ampicillin and penicillin had very strong negative linear correlation with *aac(3)-XI* ARG meaning that in the presence of phenotypic resistance (Amp, P), the genotypic resistance *aac(3)-XI* will not be present in all the isolates. Erythromycin had moderate negative linear correlation with *aac(3)-XI* ARG which means ~50 % isolates had chances of getting resistance either with erythromycin or *aac(3)-XI* ARG. Apart from this, ampicillin had very strong positive linear correlation with penicillin indicating that both the resistance will be present in a single isolate while erythromycin had moderately positive linear correlation with penicillin and ampicillin, meaning that 50% of isolates will be resistant to the above three antibiotics (Fig. 3).

3.6. Correlation matrix analysis of Gram -ve isolates

Among the Gram -ve isolates, *qnrA* ARG had a weak negative correlation with CIP, AK, PTZ, CFS, CAZ and IPM; *bla_{VIM}* had similar correlation matrix with CIP, PTZ, CFS, CAZ and IPM; *bla_{OXA-58}* with AK; *bla_{NDM}* with CIP had also similar correlation matrix. There was also linear positive correlation noticed as follows: *qnrA* gene with *bla_{OXA-58}* had a moderate linear positive correlation; *qnrB* with *bla_{OXA-48}* had a weak linear positive correlation matrix. Similarly, a weak linear positive correlation between AK and IPM, CAZ, CFS, PTZ; CIP and AK were noticed. But a strong linear positive correlation was observed between CIP and IPM, CAZ, CFS; CFS and IPM, CAZ; CAZ and IPM respectively indicating co-presence of above antibiotic resistance in a bacterium (Fig. 3).

4. Discussion

This work was taken up to establish the possible spread of antibiotic resistance through the bacteria isolated from water samples including influents, effluents, surface and ground water samples collected in and around pharmaceutical industries of south India. In this study, 48 bacterial isolates from different water samples, were used that includes Gram negative bacilli, *Klebsiella pneumoniae* (7), *Pseudomonas aeruginosa* (9), *Enterobacter aerogenes* (6), *Acinetobacter* sp. (2), *Aeromonas punctata* (2), *Ralstonia picketti* (1), *Stenotrophomonas maltophilii* (2), *Citrobacter freundii* (2) and Gram-positive cocci, *Staphylococcus aureus* (16), *Corynebacterium* sp., (1). In terms of percent prevalence among the isolates, *Staphylococcus aureus* - 28 %, *Pseudomonas aeruginosa* - 22 %, *Klebsiella pneumoniae* - 19 %, *Enterobacter aerogenes* - 12 %, *Acinetobacter* sp. - 5 %, *Citrobacter freundii* - 4 %, *Aeromonas punctata* - 3 %, *Stenotrophomonas maltophilii* - 3 %, *Corynebacterium* sp. - 2 %, *Ralstonia picketti* - 2 %, were noticed. Though this study was focused on pharmaceutical industry wastewater analysis of bacterial isolates. In the present work, comprehensive study has been performed for the first time to assess the status of phenotypic and genotypic profile of AMR along with antibiotic residue analysis in pharmaceutical industry waste water, surrounding surface and ground water. A recent study had analysed influent waste water samples from municipal of Helsinki in Finland. Upon characterizing the bacterial isolates by phenotypic and genotypic methods, they found predominantly beta lactamase resistance mediating ESBL genes.

The influent waste water samples had a total of 30 isolates (Gram positive – 11 Nos. and Gram negative – 19 Nos.) among 48 isolates used in this study but only one effluent sample had one isolate of Gram-negative type which clearly indicates that the waste water treatment plant is almost working effectively. At the same time, it is also a major concern with regard to 17 isolates obtained from environmental samples like surface and ground water samples, giving a caution about other source of contamination and spread of resistant bacteria which may lead to a major public health issue. This address the need to explore the source and spread of antibiotic resistant bacteria in the environment. One study has been carried out by (Shippingana et al., 2022) in Mysuru,

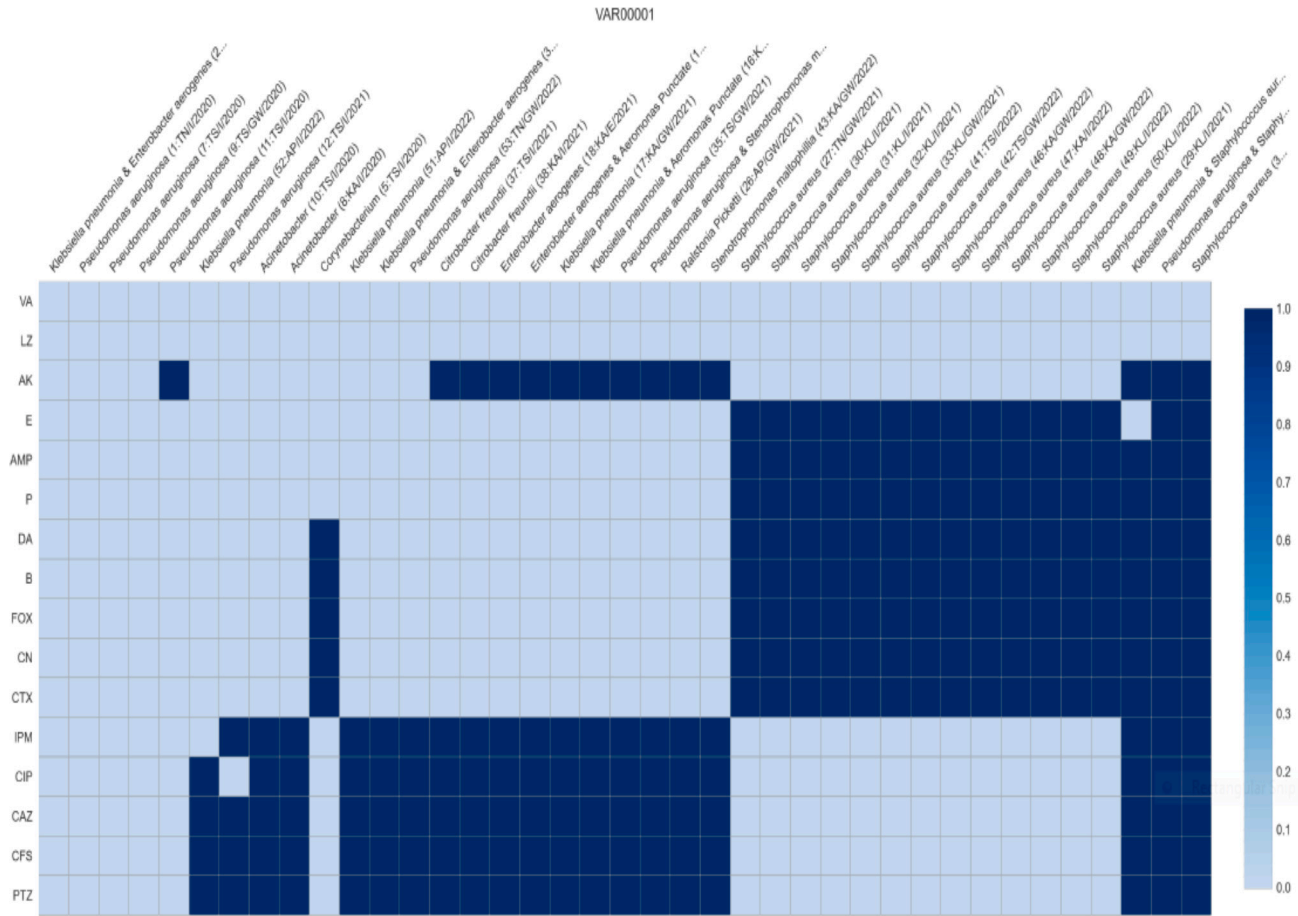


Fig. 1. Heat map analysis of phenotypic resistance among Gram -ve and Gram +ve isolates obtained from pharma industry wastewater sample.

Table 3
Phenotypic resistant/susceptible pattern of various isolates obtained from the Waste water, Surface water and Ground.

Organism/ Sample ID/ Year	Gram staining	Antibiotics (Resistant/ Sensitive)									
		E	AMP	GEN	B	VA	CD	LZ	P	CX	CTX
Corynebacterium(5:TS/1/2020)	G + ve	S	S	R	R	S	R	S	S	R	R
Staphylococcus aureus(27:TN/GW/2021)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (29:KL/1/2021)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (30:KL/1/2021)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (31:KL/1/2021)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (32:KL/1/2021)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (33:KL/GW//2021)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (34:KL/GW/2021)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (36:TS/1/2021)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (41:TS/1/2022)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus(42:TS/GW/2022)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (45:KA/1/2022)	G + ve	S	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (46:KA/GW/2022)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (47:KA/1/2022)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (48:KA/GW/2022)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (49:KL/1/2022)	G + ve	R	R	R	R	S	R	S	R	R	R
Staphylococcus aureus (50:KL/1/2022)	G + ve	R	R	R	R	S	R	S	R	R	R

Note: 54:AP/1/2022, 55:TN/E/2022, 4:TN/E/2020, 6:TS/E/2020, 13:TN/E/2021, 14:TN/E/2021, 19:TS/E/2021, 20:TS/1/2021, 21:KA/E/2021, 22:KA/E/2021, 23:KA/E/2021, 24:AP/E/2021, 25:AP/E/2021, 28:KL/E/2021, 39:KL/E/2022, 40:TS/GW/2022. All these effluent samples were negative for isolation of any bacteria. S – Sensitive, R - Resistant.

Imipenem (IPM); Ceftazidime (CAZ); Ceftaperazone/Sulbactam (CFS); Piperacillin/Tazobactam (PTZ); Amikacin (AK); Ciprofloxacin (CIP); Erythromycin (E); Ampicillin (AMP); Gentamicin (GEN); Bacitracin (B); Vancomycin (VA); Clindamycin (CD); Linezolid (LZ); Penicillin (P); Cefoxitin (CX); Cefotaxime (CTX).

India. It established the environmental contamination of four medicinal compounds including ceftriaxone in municipal wastewater treatment plant effluent. In another study conducted in Nigeria, it was revealed that the release of wastewater from slaughterhouses directly into water

sources without prior treatment, has led to significant health hazards due to multidrug resistant bacterial population contaminating the environment (Iroha et al., 2016). A high resistance to these β-lactam antibiotics is expected, given that ampicillin is among the frequently

Table 4

Genotypic resistance pattern of various Gram –ve isolates obtained from the Waste water, Surface water and Ground water samples.

Organism/ Sample ID/ Year	Gram staining	PCR Results and Resistant Antibiotics with Base pairs											
		qnrA (516)	qnrB (489)	mec (310)	bla _{VIM} (196)	bla _{OXA-58} (843)	bla _{OXA-22} (769)	aadB (500)	qepA (580)	aac(6)- lb-cr (482)	aac(3)- XI (400)	bla _{NDM} (621)	bla _{OXA-48} (438)
<i>Pseudomonas aeruginosa</i> (1:TN/1/2020)	G –ve	1	0	0	1	0	0	0	0	0	0	1	0
<i>Klebsiella pneumonia</i> (2: AP/GW/2020)	G –ve	1	0	0	0	0	0	0	0	0	0	0	0
<i>Enterobacter aerogenes</i> (2: AP/GW/2020)	G –ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Klebsiella pneumonia</i> (3: TS/GW/2020)	G –ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Enterobacter aerogenes</i> (3: TS/GW/2020)	G –ve	0	1	0	0	0	0	0	0	0	0	0	0
<i>Pseudomonas aeruginosa</i> (7: TS/1/2020)	G –ve	0	0	0	1	0	0	0	0	0	0	0	0
<i>Acinetobacter</i> (8:KA/1/2020)	G –ve	1	0	0	0	1	0	0	0	0	0	0	0
<i>Pseudomonas aeruginosa</i> (9:TS/GW/2020)	G –ve	1	0	0	0	0	0	0	0	0	0	0	0
<i>Acinetobacter</i> (10:TS/1/2020)	G –ve	1	0	0	0	1	0	0	0	0	0	0	0
<i>Pseudomonas aeruginosa</i> (11:TS/1/2020)	G –ve	0	0	0	1	0	0	0	0	0	0	0	0
<i>Pseudomonas aeruginosa</i> (12:TS/1/2021)	G –ve	0	0	0	0	0	0	0	0	0	0	1	0
<i>Enterobacter aerogenes</i> (15: KA/1/2021)	G –ve	0	1	0	0	0	0	0	0	0	0	0	0
<i>Aeromonas Punctate</i> (15: KA/1/2021)	G –ve	0	0	0	0	0	0	0	0	1	0	0	0
<i>Klebsiella pneumonia</i> (16: KA/1/2021)	G –ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Aeromonas Punctate</i> (16: KA/1/2021)	G –ve	0	0	0	0	0	0	0	0	1	0	0	0
<i>Klebsiella pneumonia</i> (17: KA/GW/2021)	G –ve	0	0	0	0	0	0	0	0	0	0	1	0
<i>Enterobacter aerogenes</i> (18: KA/E/2021)	G –ve	0	1	0	0	0	0	0	0	0	0	0	1
<i>Ralstonia Picketti</i> (26:AP/ GW/2021)	G –ve	0	0	0	0	0	1	0	0	0	0	0	0
<i>Pseudomonas aeruginosa</i> (34:KL/GW/2021)	G –ve	0	0	0	1	0	0	0	0	0	0	0	0
<i>Pseudomonas aeruginosa</i> (35:TS/GW/2021)	G –ve	0	0		0	0	0	0	0	0	0	0	0
<i>Enterobacter aerogenes</i> (36: TS/1/2021)	G –ve	0	1	0	0	0	0	0	0	0	0	0	0
<i>Citrobacter freundii</i> (37:TS/ I/2021)	G –ve	0	0	0	0	0	0	0	1	0	0	0	0
<i>Citrobacter freundii</i> (38: KA/1/2021)	G –ve	0	0	0	0	0	0	0	1	0	0	0	0
<i>Enterobacter aerogenes</i> (38: KA/1/2021)	G –ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Stenotrophomonas maltophilia</i> (43:KA/ GW/2022)	G –ve	0	0	0	0	0	0	1	0	0	0	0	0
<i>Stenotrophomonas maltophilia</i> (44:KA/1/ 2022)	G –ve	0	0	0	0	0	0	1	0	0	0	0	0
<i>Pseudomonas aeruginosa</i> (44:KA/1/2022)	G –ve	0	0	0	1	0	0	0	0	0	0	0	0
<i>Klebsiella pneumonia</i> (45: KA/1/2022)	G –ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Klebsiella pneumonia</i> (51: AP/1/2022)	G –ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Klebsiella pneumonia</i> (52: AP/1/2022)	G –ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Pseudomonas aeruginosa</i> (53:TN/GW/2022)	G –ve	0	0	0	1	0	0	0	0	0	0	0	0

The percentage of resistance means that the proportion of identified isolates that are resistant to the particular antibiotic (Gram +ve or Gram -ve) drug. Similarly, percentage of sensitivity means the proportion of the identified isolates that are sensitive to the particular antibiotic (Gram +ve or Gram -ve) drug.

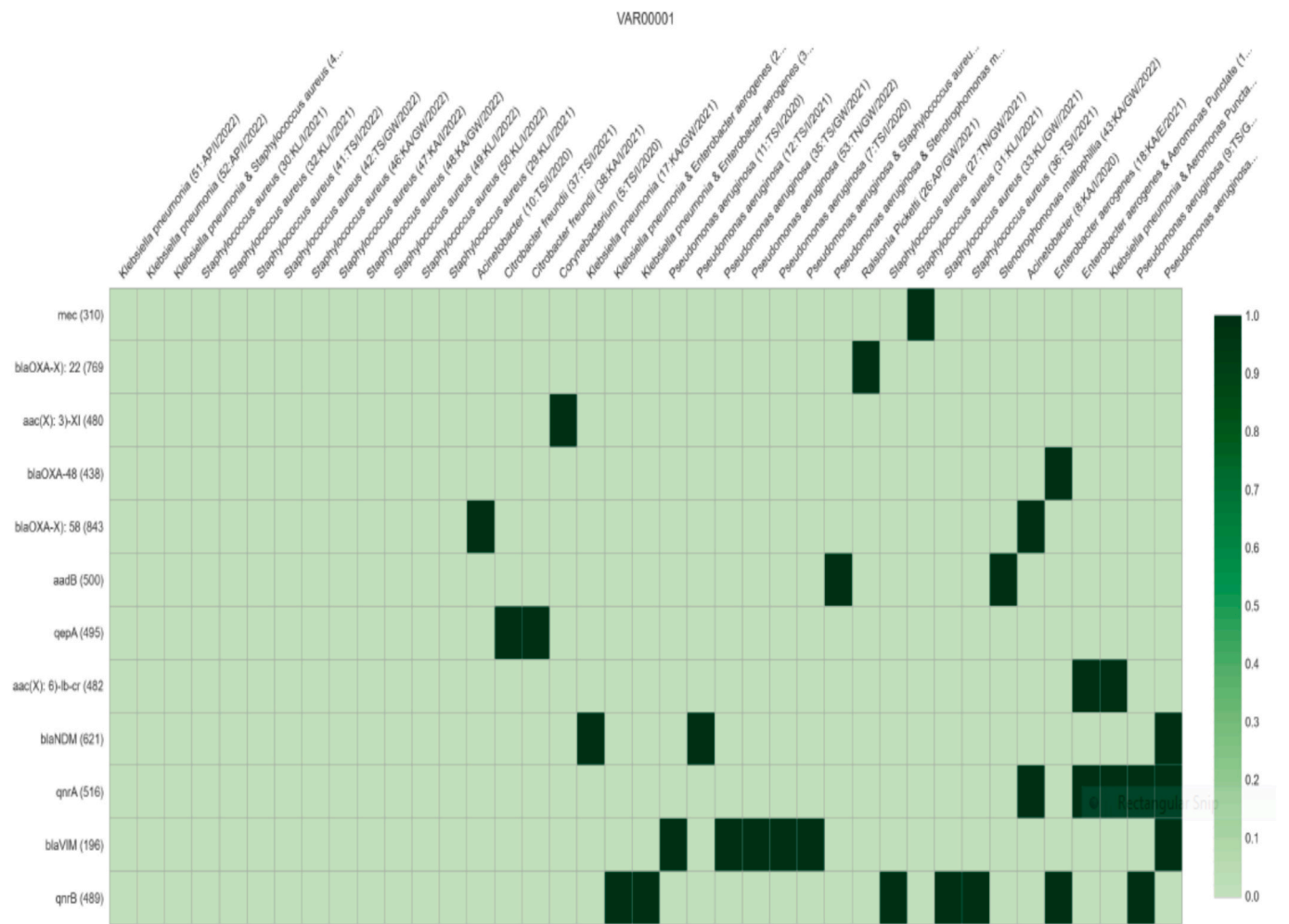


Fig. 2. Heat map analysis of genotypic resistance among Gram -ve and Gram +ve isolates obtained from pharma industry wastewater samples.

prescribed antibiotics for treating infections in both humans and animals. Moreover, resistance to penicillin in *Staphylococcus aureus* has been documented since the 1960s. Additionally, ampicillin-resistant isolates including MRSA may exhibit cross-selection for resistance to other beta-lactams (Chambers et al., 2009).

Another supporting evidence of environmental contamination was evidenced by phenotypic analysis of bacterial isolates from water samples collected after treatment in a study conducted in Mafikeng in the North West Province of South Africa. This study identified *P. aeruginosa* in both drinking and surface waters. However, in contrast to our results, a study in Alice, Eastern Cape, South Africa, analyzing wastewater samples, reported a lower occurrence rate of 11.1 %. This difference is likely attributed to variations in water purification treatment processes, or it can be inferred that wastewater treatment plants (WWTP) may not completely eradicate bacteria, especially multidrug-resistant (MDR) strains. These resilient organisms might withstand treatment processes, contributing to the transmission and spread of antimicrobial resistance (Mulamattathil et al., 2014). But in this study, only one *Enterobacter aerogenes* (18: KA/E/2021) isolate was obtained exhibiting resistance to all antibiotics tested in this study indicating the effectiveness of the wastewater treatment process.

The emergence of bacterial resistance to most frequently used antibiotics/drugs is of immense medical significance due to the public health hazards. Multiple investigations in polluted and non-polluted areas have confirmed the occurrence of antibiotic resistant microorganisms (Bielen et al., 2017). Genetic flexibilities have enabled them to survive in a wide range of environments due to their capacity to obtain

and transfer drug resistance. Aquatic environments promote gene transfer among strains of bacteria (Schages et al., 2020). In this study also, ARGs transfer through water sources in the environment was established even though the source of antibiotic resistant bacteria development is not known. This needs to be explored to find out the source of development and spread of AMR.

In this study, forty-eight bacterial isolates of Gram positive and Gram-negative types were tested for their antibiotic susceptibility and resistance against sixteen antibiotics (Table 6).

All the Gram-positive isolates tested had phenotypic resistance against cephalosporins, lincosamide, polypeptide antibiotics, aminoglycoside, and penicillin and macrolide group of antibiotics. Glycopeptide and oxazolidinone group of antibiotics were the only one to which all strains were susceptible. Among the Gram-negatives, 24 out of 31 isolates were resistant due to ESBL. Fluroquinolones and aminoglycoside resistance.

mediating genes occurred in 7 isolates. Whereas earlier reports had shown higher prevalence of multidrug resistant bacteria because the samples were sourced from common treatment plant compared with this study which focused on few pharma industries only. *Acinetobacter* sp., *Klebsiella pneumoniae*, *Proteus mirabilis*, *Enterobacter* sp., and *Bacillus* sp. estimated for the bulk of the 254 bacterial strains that were isolated; 218 (85.8 %) of these isolates were multidrug resistant (Obayiuwana et al., 2021).The resistance patterns of the isolates indicated a resistance rate of 63.9 % to second-generation cephalosporins (ceftazidime) and 55.6 % to third-generation cephalosporins (cefepime), while displaying lower resistance to aminoglycosides. The resistance pattern observed for

Table 5

Genotypic resistance pattern of various Gram +ve isolates obtained from the Waste water, Surface water and Ground water samples.

Organism/ Sample ID/ Year	Gram staining	PCR Results and Resistant Antibiotics with Base pairs											
		qnrA (516)	qnrB (489)	mec (310)	bla _{VIM} (196)	bla _{OXA-58} (843)	bla _{OXA-22} (769)	aadB (500)	qepA (580)	aac(6)- lb-cr (482)	aac(3)- XI (400)	bla _{NDM} (621)	bla _{OXA-48} (438)
<i>Corynebacterium</i> (5:TS/ 1/2020)	G + ve	0	0	0	0	0	0	0	0	0	1	0	0
<i>Staphylococcus aureus</i> (27:TN/GW/2021)	G + ve	0	1	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (29:KL/1/2021)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (30:KL/1/2021)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (31:KL/1/2021)	G + ve	0	0	1	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (32:KL/1/2021)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (33:KL/GW//2021)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (34:KL/GW/2021)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (36:TS/1/2021)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (41:TS/1/2022)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (42:TS/GW/2022)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (45:KA/1/2022)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (46:KA/GW/2022)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (47:KA/1/2022)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (48:KA/GW/2022)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (49:KL/1/2022)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0
<i>Staphylococcus aureus</i> (50:KL/1/2022)	G + ve	0	0	0	0	0	0	0	0	0	0	0	0

Note: 0 – ARG (Antibiotic Resistance Genes) absent, 1 –ARG present.

cephalosporins in the current study aligns with the findings of a previous report but is lower when compared to another study that reported 100 % resistance rate. However, in yet another study, it was documented lower resistance rates of 6.9 % and 17 % to ceftazidime and cefepime, respectively (Hosu et al., 2021).

With regard to genotypic resistance pattern, among the 17 Gram-positive isolates, only 3 isolates (two influent and one ground water samples) had ARGs for aminoglycoside, fluroquinolones and beta lactam groups. Though the above group's antibiotic resistance was phenotypically positive and the genotypic expression of those groups was not evident. They possible harbored other resistance mediator. Among the Gram-negative isolates, 10.

out of 12 ARGs were present in all 31 isolates and as expectedly, ARG viz. *aac* (3)-*XI* were not present in any of the isolates. This indicates that the beta lactam, fluroquinolones, quinolones and aminoglycosides groups ARGs were prevalent among Gram-negative bacteria. Normally, pharmaceutical industries liquid waste tends to discharge huge varieties of ARGs compared to common sewage and this was proved earlier by Tao et al. (2016). where they have observed the diversity of ARGs in pharmaceutical wastewater treatment plant aerobic sludge (153 subtypes) was higher than that in sewage treatment plant aerobic sludge (118 subtypes). A total of 12 antibiotics were checked for their effectiveness against the isolates. From this study along with earlier reports, it has been well established that the waste water treatment is an essential process to prevent or reduce the antimicrobial resistance development and spread in the environment (Davies et al., 2010; Shivakumaraswamy et al., 2019; Silva-Santana et al., 2021). These reports have shown higher level of phenotypic and genotypic resistance development and spread in the absence of wastewater treatment plant in the environment.

The correlation matrix analysis among Gram positive isolates revealed that in the presence of beta lactam groups resistance, aminoglycoside and fluroquinolones resistance were not present. Similarly, macrolides resistance never co-existed with aminoglycosides and fluroquinolones. But, all beta lactam group antibiotics resistance was present along with macrolides resistance. Similarly, in Gram-negative isolates, fluroquinolone ARGs never co-existed with beta lactam group antibiotics and beta lactam ORGs with aminoglycoside and other quinolone groups. But, in this study, a strong correlation was shown that some beta lactam and fluroquinolone antibiotics resistance were co-existed; there were also evidences for the co-existence of fluroquinolone, beta lactam group ARGs with beta lactams, fluroquinolones and aminoglycoside antibiotic groups (Shin et al., 2022; Akanbi et al., 2017). Previous study demonstrated that genotypic factors play a significant role in influencing phenotypic antibiotic resistance (Caball et al., 2022).

This study had disclosed considerable prevalence of resistance toward diverse group of antibiotics, particularly aminoglycosides, macrolides, fluroquinolones, cephalosporin and β -Lactamase, which was established by the presence of *bla*_{OXA-58}, *bla*_{OXA-22}, *qnrA*, *qnrB*, *aac*(6)-*Ib-cr*, *aac* (3)-*XI*, *mec A*, *qepA*, *aadB*, *bla*_{VIM}, *bla*_{OXA-48} and *bla*_{NDM} antibiotic resistant genes, respectively. The phenotypic resistance profile of bacterial isolates was confirmed by the genotypic testing. The variations encountered between phenotypic and genotypic antibiotic resistance is attributed to the resistance expression level of diverse genes in the bacteria.

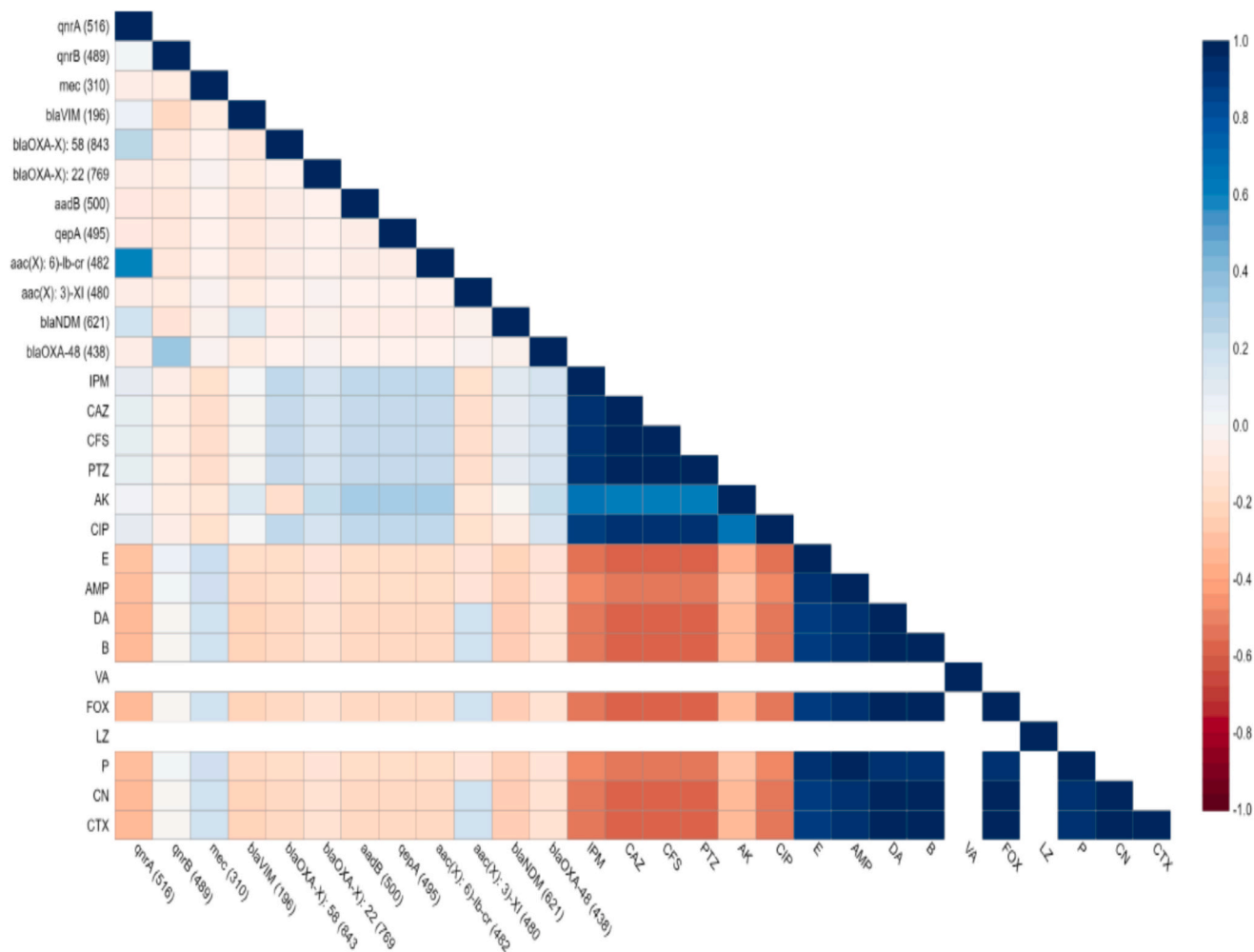


Fig. 3. Correlation matrix between the phenotypic and genotypic resistance among Gram+ve and Gram-ve isolates obtained from pharma industry waste-water sample.

Table 6
Phenotypic resistance/susceptible profile of waste water, surface water and ground water.

Gram –ve antibiotics	Percentage of resistance/susceptible	Gram + ve antibiotics	Percentage of resistance/susceptible
Imipenem (IPM)	17/16	Erythromycin (E)	11/5
Ceftazidime (CAZ)	17/14	Ampicillin (AMP)	12/2
Ceftaperazone/Sulbactam (CFS)	18/14	Gentamicin (GEN)	13/0
Piperacillin/Tazobactam (PTZ)	18/14	Bacitracin (B)	13/0
Amikacin (AK)	13/26	Vancomycin (VA)	0/44
		Clindamycin (CD)	13/0
		Linezolid (LZ)	0/44
Ciprofloxacin (CIP)	17/16	Penicillin (P)	12/2
		Cefoxitin (CX)	13/0
		Cefotaxime (CTX)	13/0

5. Conclusions

Forty-eight isolates from of 55 samples sourced from pharma industries of south India were subjected to phenotypic and genotypic antibiotic resistance analysis. Among these, 30 isolates were from influent samples, one from effluent and 17 isolates from environmental water samples. Among the bacterial isolates, 100 % of the Gram-positive isolates were phenotypically resistant to cephalosporins, lincosamide, polypeptide antibiotics, aminoglycosides, penicillin groups and macrolide group of antibiotics whereas 77 % Gram-negative isolates were phenotypically resistant to extended spectrum of beta lactamases, fluroquinolones and aminoglycoside group of antibiotics. In the genotypic antibiotic resistance gene analysis, among the 17 Gram-positive isolates, only 3 isolates had ARGs of aminoglycoside, fluroquinolones and beta lactam groups. In the Gram-negative isolates, 10 out of 12 ARGs were present in all 31 isolates which confirmed the beta lactam, fluroquinolones and aminoglycosides group's ARGs presence among Gram-negative isolates. The correlation matrix analysis showed that co-existence of resistance among penicillin and macrolide group of ARGs while penicillin group of antibiotic resistance were present along with aminoglycoside and fluroquinolone antibiotic groups. In Gram-negative isolates, evidences were noticed for the co-existence of some fluroquinolone, beta lactam group ARGs with other beta lactams, fluroquinolones and aminoglycoside antibiotic groups. By this study, it was proved that the waste water treatment plant in every pharmaceutical company is essential to stop the spread of antibiotic resistant bacterial

strain development and further spread to the environmental water sources which lead to public health hazards. Further, it was also found that antibiotic resistant bacterial strains present in the environmental water sample (surface and ground water source), needs to be investigated to find out the source of antibiotics spillage so that suitable remedial measures can be taken up. This type of study is highly essential to check the efficacy of the wastewater treatment. In spite of treatment, the effluents are harboring antimicrobial resistant bacteria which easily contaminate the water resources meant for human as well as livestock community leading to “one health” issue.

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CRediT authorship contribution statement

Sravani Kamatham: Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Manoharan Seeralan:** Software, Methodology, Formal analysis. **Uma Sekar:** Resources, Investigation, Formal analysis. **Sujatha Kuppasamy:** Supervision, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All the research data are included in the manuscript and supplementary material.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.meegid.2024.105670>.

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